

DSS Components



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- **Data Management Subsystem**
- **Model Management Subsystem**
- **User Interface Subsystem**
- **Knowledge-Based Subsystem**

Data Management Subsystem

- The Data Management Subsystem is the foundation. It's the part that handles all the data, acting as the bridge between the raw information and the analysis.
- The data management subsystem is responsible for **storing, retrieving, and managing the data** needed for decision support.
- It serves as a **repository for data** relevant to the decision-making process.
- It provides a way to **efficiently access and manipulate data**, allowing the user to get the information they need without having to manually search through multiple sources.

Where does the Data come from?

- A DSS needs access to a variety of data sources to be effective. These sources can be:
 - **Internal Data:** This is data generated within the organization.
 - **External Data:** This is data from outside the organization.
- The data management subsystem is crucial for integrating these different sources into a single, usable format.

Database Management System (DBMS)

- A **Database Management System (DBMS)** is the software that manages the database itself. It's a key part of the data management subsystem.
- Roles:
 - Data Storage: Physically stores the data in a structured way.
 - Data Retrieval: Provides a language (like SQL) for retrieving specific data.
 - Data Integrity: Ensures the data is accurate, consistent, and reliable.
 - Security: Controls who can access and modify the data.

Database vs Data Warehouse

- A **database** is typically used for day-to-day transactions and operations. It's optimized for fast updates and retrieval of small pieces of information.
- A **data warehouse** is a special type of database used for DSS. It's a large, consolidated repository of data from many different sources, optimized for complex queries and analysis.

Data Flow

1. **Data Sources:** Data is collected from both internal and external sources.
2. **ETL (Extract, Transform, Load):** Data is extracted, cleaned, and loaded into the data warehouse.
3. **Data Warehouse/Database:** The data is stored in a way that is optimized for analysis.
4. **DSS Data Management Subsystem:** The system retrieves the necessary data from the data warehouse.
5. **DSS Analysis:** The data is sent to the model management and user interface subsystems for analysis and presentation.

Real-World Example: Retail Inventory & Pricing DSS



A large retail chain running a data-driven Decision Support System to optimize product pricing and stocking levels. Its **data management subsystem** functions through specific interconnected parts:

The DSS Database / Data Warehouse – (centralized repository). Pulls Internal transactional data (past sales figures, current store inventory, supply costs) & external market data (competitor pricing, regional weather forecasts, seasonal consumer trends)

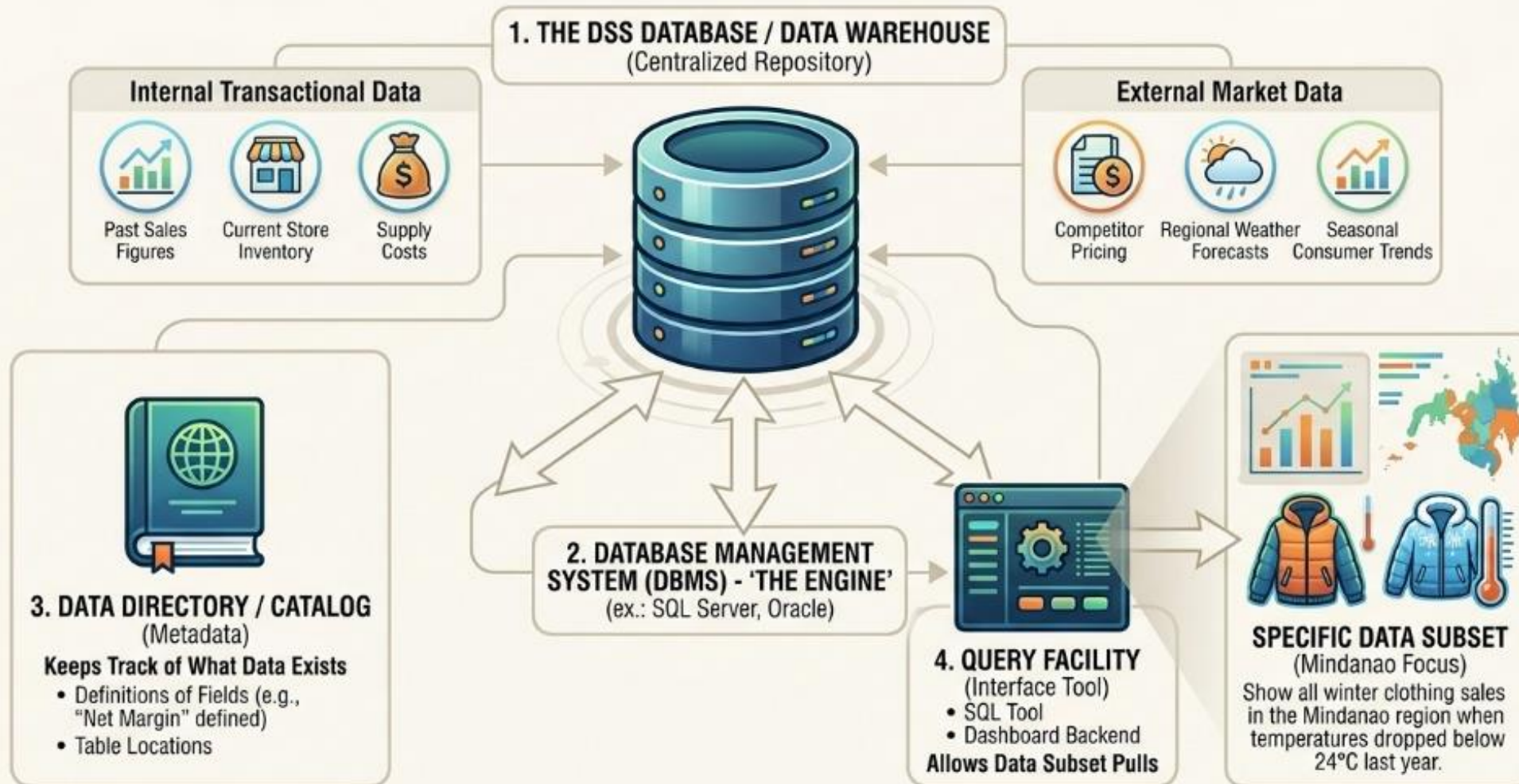
Database Management System (DBMS) – (Software like SQL Server or Oracle). Acts as the engine to compile, update, secure, and manipulate the incoming raw data.

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Query Facility - (like an SQL query tool or reporting dashboard backend). Pull specific data subsets—such as "Show all winter clothing sales in the Mindanao region when temperatures dropped below 24°C last year."

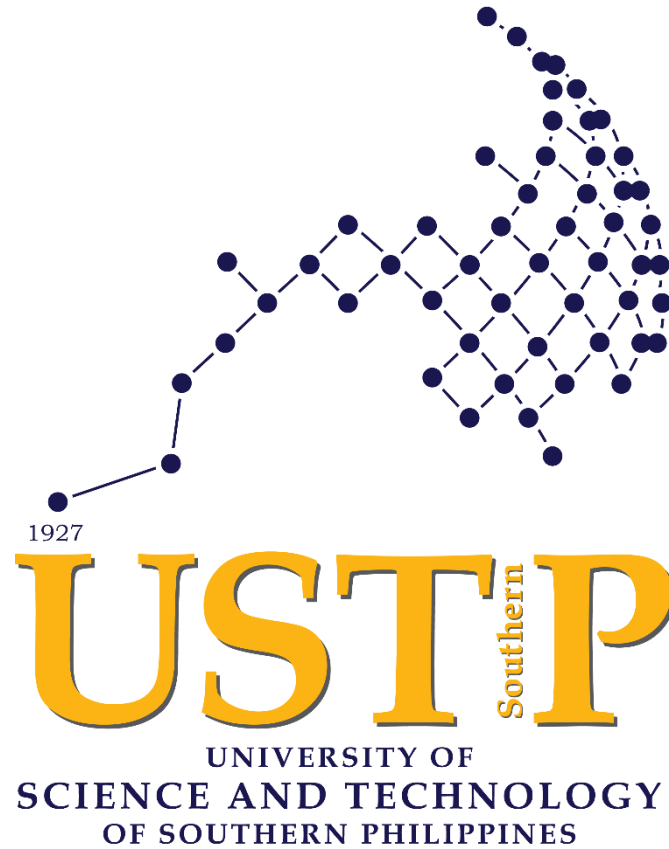
Real-World Example: Retail Inventory & Pricing DSS

RETAIL INVENTORY & PRICING DSS - DATA MANAGEMENT OVERVIEW



Discussion Questions

1. Can you identify an external data source that would be useful for a DSS helping a university with student enrollment?
2. Why is it important for the data in a DSS to be clean and accurate? What could happen if it's not?
3. How does the DBMS help ensure data integrity?



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